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AUTONOMOUS

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

MINI PROJECT REPORT

STUDENT PERFORMANCE ANALYSIS

Exploratory Data Analysis of Attendance, Study Hours, Grades, and Student Records

Submitted by

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II AI & DS - C

Under the Guidance of

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CERTIFICATE

This is to certify that the mini project report titled “Student Performance Analysis” is submitted by Sherifa S and team member Sarulatha V, from the Department of Artificial Intelligence and Data Science, SRM TRP Engineering College. The project uses data analysis techniques to explore student academic records and examine patterns involving attendance, study hours, grades, gender distribution, and relationships among numerical variables.

The report is submitted as part of the academic mini project work under the guidance of Ezhil Swanly.

ACKNOWLEDGEMENT

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We also extend our appreciation to all the faculty members of the Department of Artificial Intelligence and Data Science for their valuable inputs and encouragement. We are deeply grateful to our classmates and family members for their continuous support, cooperation, and motivation during the preparation of the healthcare dataset, data analysis notebook, and project report.

Finally, we acknowledge the healthcare datasets and spreadsheets utilized in this study, as well as the open-source Python libraries and tools that facilitated the data preprocessing, analysis, visualization, and report preparation. Their availability greatly contributed to the successful completion of this mini-project.

SHERIFA S

SARULATHA V

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ABSTRACT

Student performance analysis helps explore academic records and identify patterns that may be associated with learning outcomes. This project uses Python-based data analysis to inspect a student performance dataset, prepare the data for analysis, and create visualizations that make patterns easier to understand. The workflow includes data inspection, handling data-quality issues where required, exploratory data analysis (EDA), and visual comparison of attendance, study hours, grades, gender distribution, and numerical-variable correlations.

The project produces a cleaned dataset, a Python notebook, and visualizations. The charts include attendance versus grade, study hours versus grade, gender distribution, grade distribution, and a correlation heatmap. The report describes the analysis workflow and the purpose of each chart. Specific statistical conclusions should be drawn from the values and plots generated in the accompanying notebook rather than assumed in advance.

CHAPTER 1: INTRODUCTION

1.1 Background

Student datasets contain information that can be explored to better understand academic performance. Attendance, study hours, and assessment grades are examples of variables that can be examined using descriptive statistics and visualizations. Data analysis makes it easier to identify patterns that may not be obvious from a table of records alone.

1.2 Problem Statement

A dataset containing many student records can be difficult to interpret manually. The project addresses this by preparing the data, exploring its structure and quality, and using charts to examine distributions and relationships among selected variables.

1.3 Objectives

- Inspect the student performance dataset and understand its columns and data types.
- Identify missing values, duplicate records, and other data-quality issues.
- Prepare a cleaned dataset for analysis while avoiding unjustified changes to records.
- Explore the distribution of grades and gender in the dataset.
- Visualize relationships between attendance, study hours, and grades.
- Use a correlation heatmap to examine relationships among numerical variables.
- Document the workflow and visualizations in a reproducible project repository.

1.4 Scope

The project focuses on descriptive analysis and visualization of the available student records. It explores patterns in the variables present in the dataset. The analysis does not establish that a particular factor causes a student to achieve a particular grade, and no predictive model is claimed unless one is separately implemented and evaluated.

1.5 Expected Outcome

The expected outcome is an organized dataset, a Python notebook documenting the analysis, a set of visualizations, and a report explaining the project workflow and the purpose of the charts.

CHAPTER 2: DATASET DESCRIPTION

2.1 Dataset Overview

The project uses a student performance dataset supplied and prepared for this analysis. It contains student-related academic information, including variables used to explore attendance, study hours, grades, and gender. The exact column names and record counts should be taken from the final workbook and notebook.

2.2 Variables Used in the Analysis

Variable / Feature	Purpose	Analysis
Attendance	Represents student attendance information available in the dataset.	Compared with grades.
Study Hours	Represents recorded study time.	Compared with grades.
Grade	Represents the academic outcome or grade field in the dataset.	Distribution and relationship analysis.
Gender	Represents the gender category recorded in the dataset.	Category distribution.
Other numerical fields	Additional numeric student variables, if present.	Correlation heatmap where appropriate.

2.3 Dataset Preparation

The project keeps an uncleaned workbook and a cleaned workbook so that the preparation process can be documented. The analysis notebook should be used as the source of truth for the final record count, field names, and exact cleaning operations.

CHAPTER 3: METHODOLOGY

3.1 Project Workflow

1. Load the original student dataset into the Python environment.
2. Inspect the dataset shape, column names, data types, missing values, and duplicate records.
3. Review data-quality issues and decide how to handle them based on the data and project requirements.
4. Save the prepared dataset separately from the original dataset.
5. Perform exploratory data analysis using summaries and visualizations.
6. Create charts for attendance versus grade, study hours versus grade, gender distribution, grade distribution, and numerical correlations.

7. Interpret the charts carefully and document the findings and limitations.

3.2 Tools and Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab / Jupyter Notebook
- Microsoft Excel-compatible workbook files
- GitHub for project versioning and sharing

CHAPTER 4: DATA PREPROCESSING

4.1 Data Inspection

The first step is to inspect the dataset's structure, column names, data types, and summary information. This helps identify whether fields have been read in the correct format and whether further cleaning is required.

4.2 Missing Values and Duplicate Records

Missing values and duplicate rows should be checked before analysis. A record should not be deleted or altered automatically without considering what the value represents. Any correction or removal should be justified and documented in the notebook.

4.3 Data Type and Consistency Checks

Numerical fields such as attendance, study hours, and grade should be checked for suitable data types and plausible values. Category labels, including gender or grade categories, should be reviewed for inconsistent spelling or formatting.

4.4 Cleaned Dataset

The cleaned workbook is used for downstream analysis. The exact number of rows retained and the changes made during cleaning should be reported from the final notebook output, so the report remains consistent with the actual project files.

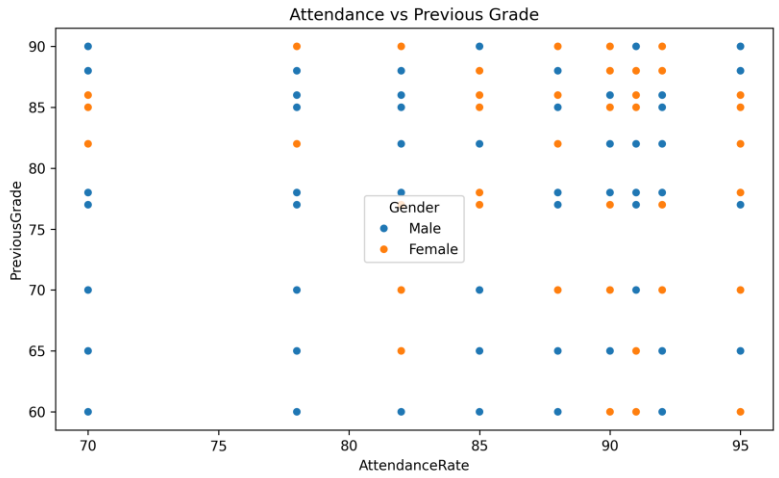
CHAPTER 5: EXPLORATORY DATA ANALYSIS

5.1 Overview

Exploratory Data Analysis (EDA) is used to summarize the dataset and explore patterns before drawing conclusions. This project uses charts to review category distributions and relationships between selected academic variables.

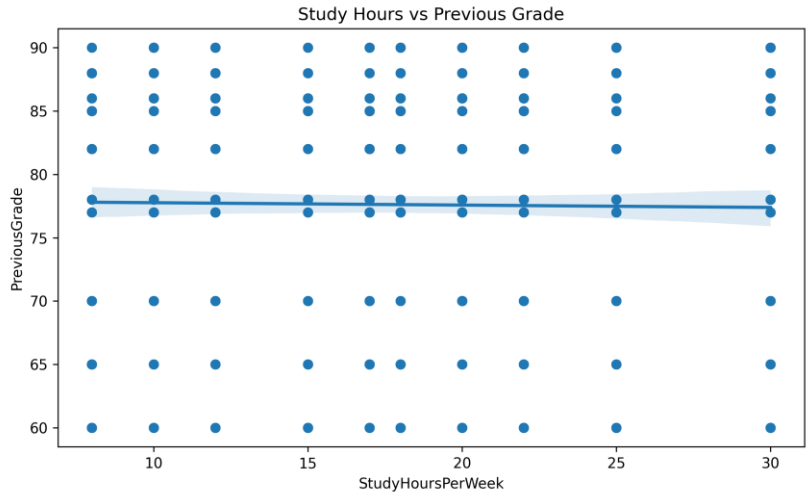
5.2 Attendance vs Grade

The attendance-versus-grade chart helps inspect whether students with different attendance levels also show different grade patterns. A visible association should be described as a relationship in the dataset, not as proof that attendance alone causes a grade outcome.



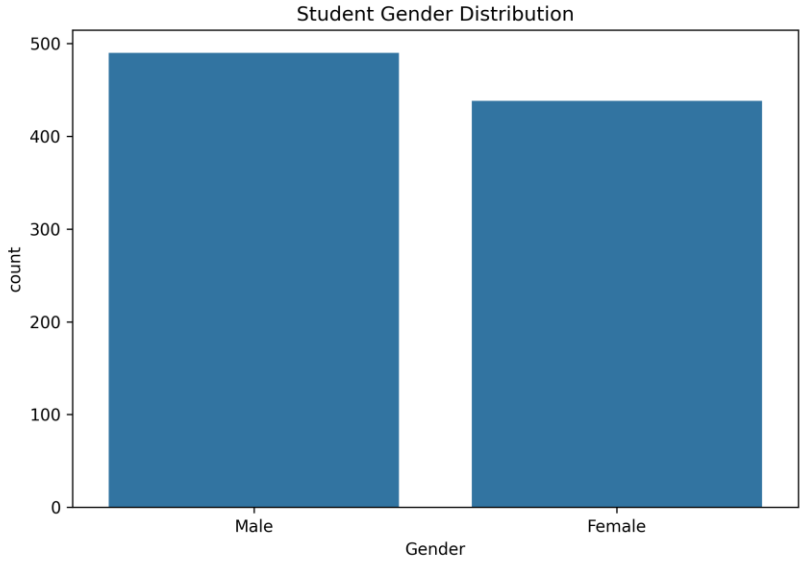
5.3 Study Hours vs Grade

The study-hours-versus-grade chart compares recorded study time with grades. It can help identify broad trends, variation, or unusual observations. The interpretation should be based on the actual plotted data.



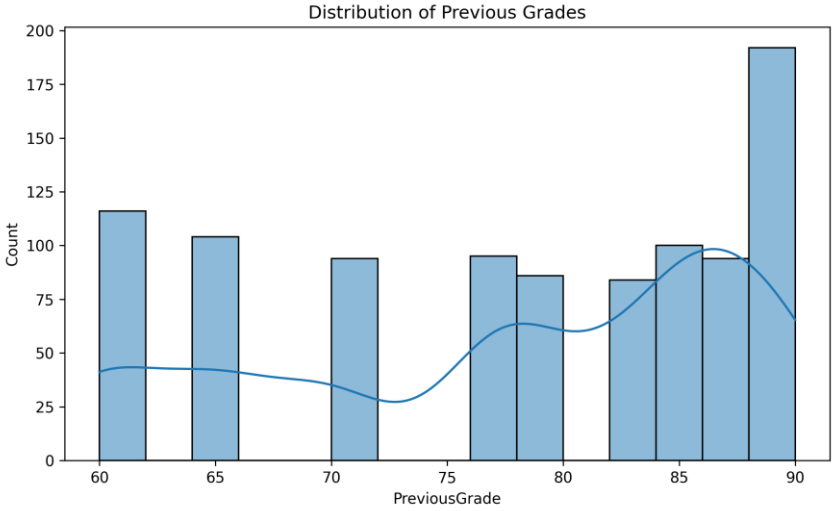
5.4 Gender Distribution

The gender distribution chart summarizes the number of records in each gender category represented in the dataset. It provides a simple view of the composition of the records and should not be interpreted as a statement about a wider population unless the dataset is representative.



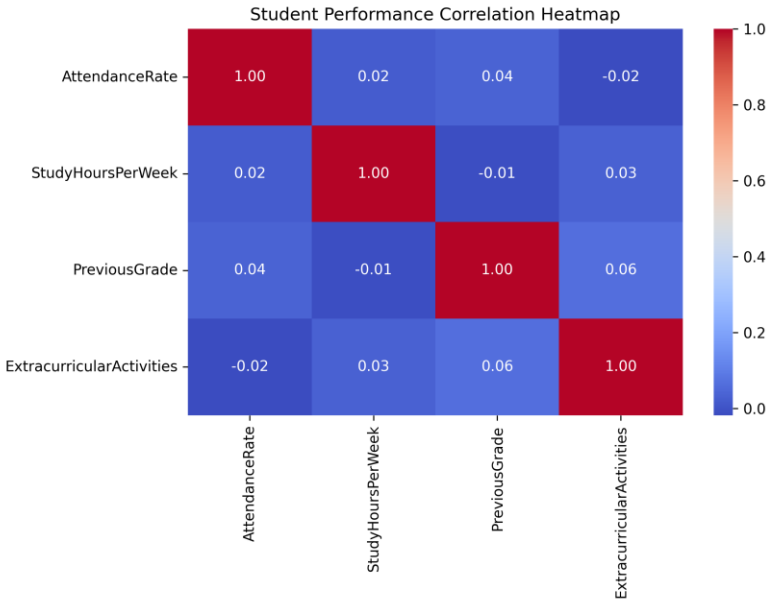
5.5 Grade Distribution

The grade distribution chart shows how records are spread across the available grade values or categories. It helps identify which grades occur more or less often in the dataset.



5.6 Correlation Heatmap

The correlation heatmap displays correlation values among selected numerical variables. Correlation indicates the strength and direction of a linear relationship; it does not establish causation. Correlation results also depend on the fields included and the quality of the data.



5.7 Visualizations Included

- Figure 5.1: Attendance vs Grade — explores attendance and grade patterns.
- Figure 5.2: Study Hours vs Grade — explores study hours and grade patterns.
- Figure 5.3: Gender Distribution — displays the distribution of gender categories.
- Figure 5.4: Grade Distribution — displays the distribution of grades.
- Figure 5.5: Correlation Heatmap — summarizes correlations among numerical variables.

CHAPTER 6: FINDINGS AND DISCUSSION

6.1 Interpreting the Results

The charts provide complementary views of the student dataset: distributions describe how values are spread, scatter plots help inspect relationships between variables, and a correlation heatmap summarizes pairwise linear relationships among numerical fields.

6.2 Key Discussion Points

- Use the attendance-versus-grade chart to describe the pattern visible in the plotted records.
- Use the study-hours-versus-grade chart to discuss the observed relationship and any variation.
- Use the grade and gender distributions to describe the composition of the dataset.
- Use the correlation heatmap to identify numerical relationships that may merit further investigation.
- Confirm any numerical claims using the notebook outputs before including them in a final presentation.

6.3 Practical Use

The analysis can support an initial understanding of student records and help identify questions for further investigation. It should be used as a descriptive aid rather than as a sole basis for decisions about individual students.

CHAPTER 7: LIMITATIONS AND FUTURE WORK

7.1 Limitations

- The analysis is limited to the variables available in the supplied dataset.
- Observed relationships do not prove cause and effect.
- The results may not generalize beyond the records included in the dataset.
- Data-quality decisions can affect the results and should be documented.
- The report does not claim a predictive model unless a model is built and evaluated separately.

7.2 Future Work

- Validate the dataset and document the exact number of records before and after cleaning.
- Add further descriptive statistics and compare groups where appropriate.
- Explore additional variables if reliable data is available.
- Evaluate a suitable predictive model only if required, using an appropriate validation method.
- Create an interactive dashboard if the project requirements call for one.

CHAPTER 8: CONCLUSION

This project demonstrates a practical workflow for student performance analysis using Python. It covers data inspection and preparation, exploratory data analysis, and visualizations for attendance, study hours, grades, gender distribution, and correlations among numerical variables. The notebook, cleaned dataset, visualizations, and report provide a structured foundation for understanding the available student records. Final numerical findings should be verified against the saved notebook and dataset.

REFERENCES

- Pandas documentation: https://pandas.pydata.org/docs/
- NumPy documentation: https://numpy.org/doc/
- Matplotlib documentation: https://matplotlib.org/stable/
- Seaborn documentation: https://seaborn.pydata.org/

APPENDIX: PROJECT FILES

The GitHub repository is organized into folders for the datasets, analysis notebook, and visualizations.

- dataset/student_performance_uncleaned.xlsx
- dataset/student_performance_cleaned.xlsx
- notebook/student_performance_analysis.ipynb
- visualizations/attendance_vs_grade.png
- visualizations/correlation_heatmap.png
- visualizations/gender_distribution.png
- visualizations/grade_distribution.png
- visualizations/study_hours_vs_grade.png
- README.md

GitHub Repository: https://github.com/sherifasyed/student-performance-analysis